Alivia G. Nytko

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• https://anytko.github.io www.github.com/anytko

Education

Ph.D. Ecology and Evolutionary Biology — August 2022 – *Expected* December 2026
Interdisciplinary Minor in Computational Science — Current GPA 4.00/4.00
The University of Tennessee, Knoxville, TN
Advisors: Joe Bailey & Brian O'Meara

B.S. Biology – Ecology & Evolutionary Biology Concentration — May 2022
The University of Tennessee, Knoxville, TN — GPA 3.91/4.00
Thesis Advisor: Jen Schweitzer

Software Engineering Bootcamp — June 2024
The University of Tennessee, Knoxville, TN
Center for Professional Educational & Lifelong Learning

Fellowships/Scholarships ($220,500)

NSF Graduate Research Fellowship Program (GRFP)	$159,000	2022-2027
Yates Dissertation Fellowship (UTK)	$10,000	2026
Dawn and Lawrence Taylor Dissertation Award (UTK)	$5,000	2025
Volunteer Scholarship (UTK)	$27,000	2019-2022
Distinguished Tennessean Award	$4,500	2019-2022
Hope Scholarship with Merit	$15,000	2019-2022

Research Grants ($14,250)

THRIVE: Tennessee Human-AI Readiness & Innovation (declined)	$10,000	2026
UTK EEB Graduate Research Award	$2,750	2023-2024
Faculty Research Assistant Funding (FRAF)	$1,500	2021-2022

Awards ($6,250)

Ernst Mayr Award Finalist (Society of Systematic Biologists)		2025, 2026
Graduate Student Travel Award (ASN)	$500	2024
Best Progress Toward Dissertation (EEB UTK)		2025
E. Lucy Braun Award (Ecological Society of America)	$500	2024
Tom Hallam Award (EEB UTK)	$500	2024
Ray Holton Plant Science Award (BCMB UTK)	$2,500	2023
Global Catalyst Student International Travel Grant (UTK)	$850	2023
Tennessee Plant Research Center Travel Fellowship	$500	2023
Graduate Student Senate (GSS) Travel Award	$1,200	2023
EEB GREBE Travel Award	$100	2023

Best Science Communication at EEB Undergrad Symposium	$100	2022
Volunteer of Distinction – Extraordinary Academic Achievement		2022
Volunteer of Distinction – Extraordinary Professional Promise		2022
Best Student Presentation Award at FEWSUS International Symposium		2021

Publications

1. **A.G. Nytko**, J.K. Senior, R.C. Wooliver, J. O'Reilly-Wapstra, J.A. Schweitzer, J. Bailey. An evolutionary case for plant rarity: *Eucalyptus* as a model system. *Ecology and Evolution* (2024), https://doi.org/10.1002/ece3.11440.

2. **A.G. Nytko**, A.M. Hord, J.K. Senior, J. O'Reilly-Wapstra, J.A. Schweitzer, J. Bailey. Evolution of rarity and phylogeny determine above- and belowground biomass in plant-plant interactions. *PLOS ONE* (2024), https://doi.org/10.1371/journal.pone.0294839.

3. N. Cooper, & **BES Data and Code Hackathon Group**. Data- and code-archiving in the British Ecological Society journals: Present status and recommendations for future improvements. *Methods in Ecology and Evolution* (2026), https://doi.org/10.1111/2041-210x.70338.

4. S.J. Love, J.D. Edwards, C.N. Barnes, T.W. d'Entremont, A.M. Hord, **A.G. Nytko**, N.B. Sero, S.N. Kivlin, J.K. Bailey. Identifying research targets and risks in climate change studies to break barriers to broad inference. *PLOS Climate* (2023), https://doi.org/10.1371/journal.pclm.0000320.

In Review

5. **A.G. Nytko**, B.C. O'Meara, J.K. Bailey. A general model of discretized rarity and its sub-models.

6. **A.G. Nytko**, J.K. Bailey, J. Beaulieu, M. Muñoz, C. Román-Palacios, B.C. O'Meara. Tree thinking in ecology: What we gain by engaging with evolutionary history. Invited perspective for *Methods in Ecology and Evolution*.

7. **A.G. Nytko**, B.C. O'Meara, J.K. Bailey. The tortoise and the hare: Range dynamics of North American plant species.

8. A.M. Hord, **A.G. Nytko**, L. Politano, J.K. Bailey, J.A. Schweitzer. How to use a biogeographic, 'genes to ecosystems' framework for mechanistic understanding of the assembly and disassembly of terrestrial forest ecosystems.

In Preparation

9. **A.G. Nytko**, J.K, Bailey, B.C. O'Meara. Evolutionary diversification and turnover help explain the persistence of multidimensional rarity.
10. **A.G. Nytko**, B.C. O'Meara, J.K. Bailey. Multidimensional rarity captures non-additive variation in plant distribution dynamics and phenology.

Scientific Software

***GDRarity* R package (DOI: 10.5281/zenodo.17214385)**

- https://github.com/anytko/GDRarity

- Calculates and compares species' geographic, functional, and phylogenetic dimensions of rarity at local and global scales in a general model of discretized rarity.

***Ecospat* Python package and web application**

- https://pypi.org/project/ecospat/
- https://huggingface.co/spaces/anytko/ecospat
- Maps and identifies historical and contemporary range edges, assigning range movement dynamics to species distributions.
- Quantifies northward movement rates and changes in population density through time.
- Estimates propagule pressure and predicts the persistence of individuals.

EcoSMART: Statistics, Modeling, Analysis of Ranges and Traits

- https://anytko.shinyapps.io/ecosmart/
- Maps and calculates species native and invasive ranges.
- Provides animations representing species occurrences through time.
- Displays connected populations through time and estimates dispersal rates.

Presentations and Talks

*Contributed presentations (presenter indicated with *):*

1. **A.G. Nytko*** and B.C. O'Meara. Rare today, common tomorrow: Evolutionary transitions and non-equilibrium dynamics shape species rarity. Evolution, Cleveland, OH. 2026

2. **A.G. Nytko*** and B.C. O'Meara. Rare today, common tomorrow: Evolutionary transitions and non-equilibrium dynamics shape species rarity. Evolution, online (Mayr Award finalist). 2026

3. **A.G. Nytko***, B.C. O'Meara, and J.K. Bailey. A global model of discretized rarity. Ecological Society of America, Baltimore, MD. 2025

4. **A.G. Nytko***, J.K. Bailey, and B.C. O'Meara. Why rarity is common. The evolution of eco-evolutionary rarity in plants. Evolution, Athens, GA. 2025

5. **A.G. Nytko***, J.K. Bailey, and B.C. O'Meara. Why rarity is common. The evolution of eco-evolutionary rarity in plants. Evolution, online (Mayr Award finalist). 2025

*Selected contributed posters (presenter indicated with *):*

1. **A.G. Nytko***, J. Senior, R. Wooliver, J. O'Reilly-Wapstra, J.A. Schweitzer, and J.K. Bailey. An evolutionary case for plant rarity. Ecological Society of America, Portland, OR. (E. Lucy Braun Award winner). 2023

2. **A.G. Nytko***, S. Turner, J.A. Schweitzer, and J.K. Bailey. Fungal communities may mediate aboveground biomass in increasingly diverse plant communities. 26th Annual Exhibition of Undergraduate Research and Creative Achievement (EUReKA). Knoxville, TN. 2022

3. **A.G. Nytko***, S. Turner, J.A. Schweitzer, and J.K. Bailey. Fungal communities may mediate aboveground biomass in increasingly diverse plant communities. 6th Annual EEB Undergraduate Research Symposium. Knoxville, TN. 2022

4. **A.G. Nytko***, Q. He, and C. Swanson. Comparative study of the microbiome of carbon filters and the microbiome of their subsequent drinking water. University of Tennessee Summer Research Scholars Symposium. Knoxville, TN. 2021

5. **A.G. Nytko*** and J. Zhuang. Perceptions of potential biosolid barrier solutions for agricultural use in Tennessee. 25th Annual Exhibition of Undergraduate Research and Creative Achievement (EUReKA). Knoxville, TN. 2021

6. **A.G. Nytko*** and J. Zhuang. Perceptions of potential biosolid barrier solutions for agricultural use in Tennessee. NSF Student FEWSUS International Symposium, online. 2021

Conferences, Working Groups, & Symposia Organized

Special Session Organizer and chair. Breaking barriers to incorporating phylogenetics in ecology. Ecological Society of America. Baltimore, MD. 2025

- https://anytko.github.io/breaking_barriers_phylogenetic_ecology

Workshops Attended

Shaping Imageomics Tools for Biologists – Evolution Annual Meeting 2026
Phylogenomics Software School – Evolution Annual Meeting 2025
An introduction to close range LiDAR forest ecology – BES 2023

Teaching Experience

The University of Tennessee, Knoxville, TN 2022 – 2023
Graduate Teaching Assistant
Organismal & Ecological Biology – BioLit
Ecology

The University of Tennessee, Knoxville, TN 2020 – 2021
Undergraduate Teaching Assistant
Skills of Biological Investigation Lab

Research Skills

Coding Languages: R (advanced), Python (intermediate), Bash (intermediate), JavaScript (beginner), Django (beginner), React Redux (beginner)
Ecological Niche Modeling: ArcGIS (intermediate), QGIS (intermediate), MaxEnt (intermediate), Google Earth Engine (beginner)
Version Control: Git and GitHub (advanced)
High-Performance Computing: Parallel computing, SLURM, GPU computing
Molecular Laboratory Methods: DNA extraction, PCR, Next-generation sequencing

Professional Service

Manuscript Review

Proceedings of the National Academy of Sciences (PNAS), Evolution, New Phytologist, Biodiversity and Conservation

Professional Society Service

President, Delta Alpha Pi Honors Society, University of Tennessee 2021 – 2023

- Facilitated 3rd Annual DAPi Disability Awareness Seminar

Outreach

Mycelium Committee Chair 2025 – 2026

Darwin Day Committee 2024 – 2025

- Developed a K-12 educational activity on phylogenetics for McClung Museum Darwin Day

Sigma Alpha Graduate Application Workshop

UTK College of Arts and Sciences Undergraduate Orientation

Professional Society Membership

British Ecological Society
Society of Systematic Biologists
Society for the Study of Evolution
American Society of Naturalists
Delta Alpha Pi Honors Society

References

Ph.D. Advisor: Joseph Bailey, Ph.D., UTK: jbaile29@utk.edu
Ph.D. Advisor: Brian O'Meara, Ph.D., UTK: bomeara@utk.edu
Committee Member: Jeremy Beaulieu, Ph.D., UArk: jmbeauli@uark.edu
Non-Committee Member: Enes Ozel, Ph.D., UTK: eozel@utk.edu
Peer: Ashlynn Hord, Ph.D. Candidate, UTK: ahord1@vols.utk.edu